

Problem 9.6

It is known that

$$1 + \frac{i^{(n)}}{n} = \frac{1 + \frac{i^{(4)}}{4}}{1 + \frac{i^{(5)}}{5}}.$$

Find n .

Solution

For a nominal annual interest rate convertible k times per year,

$$1 + \frac{i^{(k)}}{k} = (1 + i)^{1/k},$$

where i is the equivalent effective annual interest rate.

Therefore,

$$1 + \frac{i^{(n)}}{n} = (1 + i)^{1/n},$$
$$1 + \frac{i^{(4)}}{4} = (1 + i)^{1/4}, \quad 1 + \frac{i^{(5)}}{5} = (1 + i)^{1/5}.$$

Substituting into the given equation,

$$\begin{aligned}(1 + i)^{1/n} &= \frac{(1 + i)^{1/4}}{(1 + i)^{1/5}} \\ &= (1 + i)^{1/4 - 1/5} \\ &= (1 + i)^{1/20}.\end{aligned}$$

Thus,

$$\frac{1}{n} = \frac{1}{20},$$

so

$$\boxed{n = 20}.$$